

Brian Chien

Phone: 301-852-3355

Email: brianchien04173@gmail.com

Website: bchien21.github.io

EDUCATION

Johns Hopkins University

M.S. in Robotics

Baltimore, MD

Aug 2024 – Aug 2026

Carnegie Mellon University

B.S. in Mechanical Engineering

Pittsburgh, PA

Aug 2020 – May 2024

SKILLS

Programming Languages: Python, C++, MATLAB

Software Stack: ROS2, PyTorch, LangChain, Isaac Lab/Sim, MCP, llama.cpp, OpenCV, Git/Github (CI/CD), Docker

WORK EXPERIENCE

Hitachi America, Ltd

Artificial Intelligence Intern

Santa Clara, CA

Jan 2026 – Jun 2026

- Built an agentic AI stack for LLM-driven orchestration of a PLC-enabled factory testbed (Fischertechnik), deploying an agent on each machine to demonstrate flexible workflows and autonomous coordination.
- Developed vision-based MCP tools for agents to autonomously recover from faults without manual intervention.
- Quantized a local LLM with llama.cpp then deployed it on a Jetson AGX Orin to power the reasoning of lower-complexity agents, achieving over 50% reduction in inference time compared to cloud models.
- Engineered a ROS2 pipeline for real-time lidar SLAM of a factory-inspection Unitree quadruped.

Johns Hopkins University

Graduate Research Assistant, Autonomous Control & Exploration Lab

Baltimore, MD

Jun 2025 – Dec 2025

- Built reinforcement learning (RL) pipelines with NVIDIA Isaac Lab for a quadcopter and rover platform.
- Defined each robot's control policy in PyTorch, and leveraged the Isaac Lab pipelines to train and test both policies on goal-directed navigation via NVIDIA Isaac Sim SITL.
- Developed a ROS2 quadcopter motion planner to generate datasets of trajectories through obstacle fields; utilized this dataset to train a PyTorch diffusion policy via imitation learning (IL) for collision-free path generation.

Graduate Research Assistant, Intuitive Computing Lab

Feb 2025 – Jun 2025

- Refactored the monolithic codebase of a conversational robot into a modular design, with modules dedicated to a specific subsystem such as API calling or robot control, reducing feature development time by roughly 25%.
- Developed a multi-stage prompting pipeline to improve response duration by 50%, allowing for richer dialogue.

Ford Motor Company

Manufacturing Engineering Intern

Glendale, KY

May 2024 – Aug 2024

- Fine-tuned a machine-vision model for OCR and image classification using datasets of roughly 20 images, delivering a proof-of-concept to leadership to illustrate the viability of AI-driven anomaly detection.
- Automated plant-machine installation tracking via Excel VBA, cutting a 30-minute update process to seconds.

Carnegie Mellon University

Undergraduate Research Assistant, Biorobotics Lab

Pittsburgh, PA

Sep 2023 – Apr 2024

- Developed ROS nodes for real-time pose tracking of a surgical robot snake during live operations.
- Designed, 3D-printed, and integrated a custom medicine-delivery mechanical subsystem on the robot platform.

PROJECTS

Hierarchical Path Planning and Trajectory Optimization for a Quadcopter Platform

Nov 2025 – Dec 2025

- Developed a ROS2 service for quadcopter RRT* path planning, and co-developed a model predictive controller to output control inputs subject to constraints involving the planned path; demonstrated with Gazebo SITL.

RRT Path Planning for a UR5 6-DOF Manipulator

Oct 2024

- Implemented bidirectional RRT path-planning from scratch in C++, for a 6-DOF manipulator to generate collision-free paths into a constrained car-interior environment in roughly 5 seconds.

Automated Assistant For Drill Bit Measurement

Oct 2023 – Dec 2023

- Led a 3-person senior capstone team, and demonstrated automated drill bit measurement in under 10 seconds.
- Primarily responsible for developing Arduino firmware for motor control and sensor monitoring.